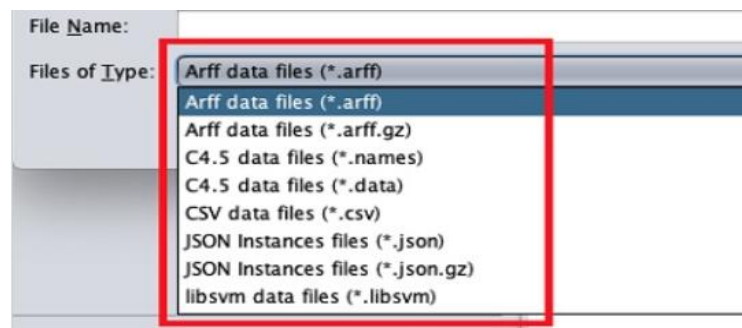


Weka - File Formats

WEKA supports a large number of file formats for the data. Here is the complete list:

- arff
- arff.gz
- bsi
- csv
- dat
- data
- json
- json.gz
- libsvm
- m
- names
- xrff
- xrff.gz



As you would notice it supports several formats including CSV and JSON. The default file type is Arff.

Arff Format

Attribute-Relation File Format (ARFF) is an ASCII text file format used for machine learning datasets, primarily designed for Weka. The file contains two sections - header and data.

- The header describes the attribute types.
- The data section contains a comma separated list of data.

- As an example for Arff format, the `Weather` data file loaded from the WEKA sample databases is shown below:

```
@relation weather.symbolic
@attribute outlook {sunny, overcast, rainy}
@attribute temperature {hot, mild, cool}
@attribute humidity {high, normal}
@attribute windy {TRUE, FALSE}
@attribute play {yes, no}
@data
sunny,hot,high,FALSE,no
sunny,hot,high,TRUE,no
overcast,hot,high,FALSE,yes
rainy,mild,high,FALSE,yes
rainy,cool,normal,FALSE,yes
rainy,cool,normal,TRUE,no
overcast,cool,normal,TRUE,yes
sunny,mild,high,FALSE,no
sunny,cool,normal,FALSE,yes
rainy,mild,normal,FALSE,yes
sunny,mild,normal,TRUE,yes
overcast,mild,high,TRUE,yes
overcast,hot,normal,FALSE,yes
rainy,mild,high,TRUE,no
```

Diagram annotations:

- Dataset name**: Points to `@relation weather.symbolic`
- Attributes**: Points to the list of attributes: `@attribute outlook {sunny, overcast, rainy}`, `@attribute temperature {hot, mild, cool}`, `@attribute humidity {high, normal}`, `@attribute windy {TRUE, FALSE}`, and `@attribute play {yes, no}`
- Target / Class variable**: Points to the `play` attribute in the data rows, which is the last field in each row.
- Data Values**: Points to the data rows starting from `@data`.

- The `@relation` tag defines the name of the database.
- The `@attribute` tag defines the attributes.
- The `@data` tag starts the list of data rows each containing the comma separated fields.

Other Formats

The Explorer can load the data in any of the earlier mentioned formats. As arff is the preferred format in WEKA, you may load the data from any format and save it to arff format for later use. After preprocessing the data, just save it to arff format for further analysis.

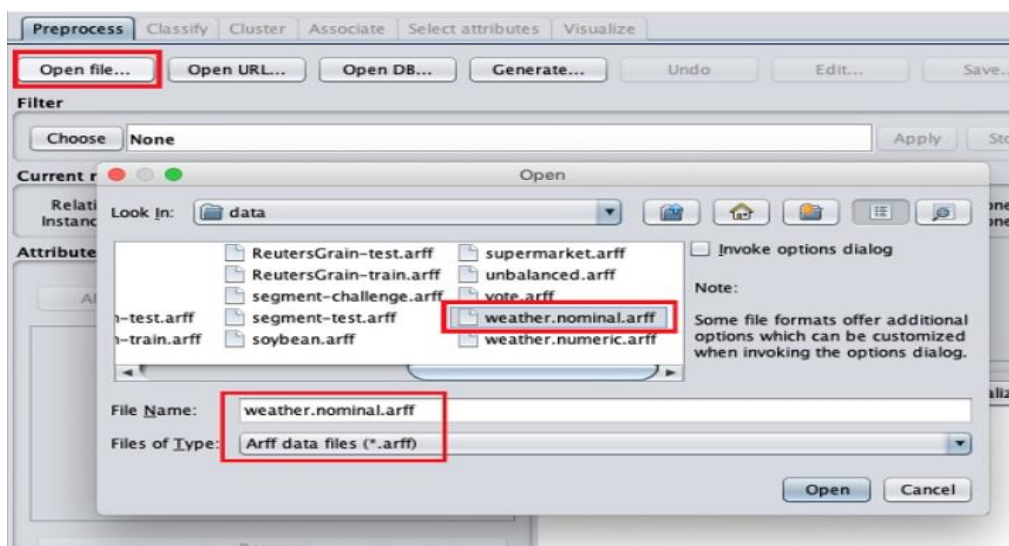
Weka - Preprocessing the Data

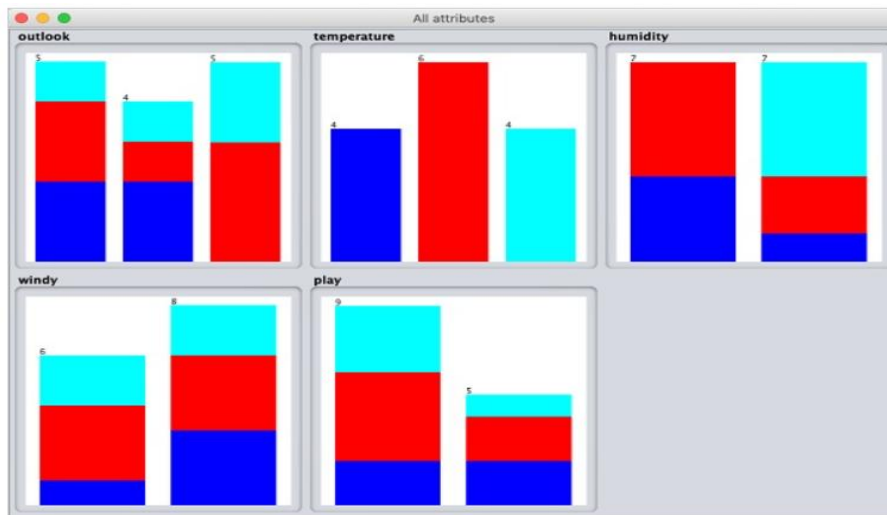
In Weka, **filters** are preprocessing tools used to modify data before applying machine learning algorithms. The key distinction is whether the **class label (target variable)** is used during the filtering process.

The data that is collected from the field contains many unwanted things that leads to wrong analysis. For example, the data may contain null fields, it may contain columns that are irrelevant to the current analysis, and so on. Thus, the data must be preprocessed to meet the requirements of the type of analysis you are seeking. This is done in the preprocessing module.

To demonstrate the available features in preprocessing, we will use the **Weather** database that is provided in the weka installation.

Using the **Open file...** ... option under the **Preprocess** tag select the **weather-nominal.arff** file.





Removing Attributes

Many a time, the data that you want to use for model building comes with many irrelevant fields. For example, the customer database may contain his mobile number which is relevant in analyzing his credit rating.

The figure shows a window titled "Attributes" with four buttons: "All", "None", "Invert", and "Pattern". Below the buttons is a table with 5 rows of attributes. The first and third rows are selected. At the bottom is a "Remove" button.

No.		Name
1	☒	outlook
2	☐	temperature
3	☒	humidity
4	☐	windy
5	☐	play

To remove Attribute/s select them and click on the **Remove** button at the bottom.

The selected attributes would be removed from the database. After you fully preprocess the data, you can save it for model building.

Applying Filters

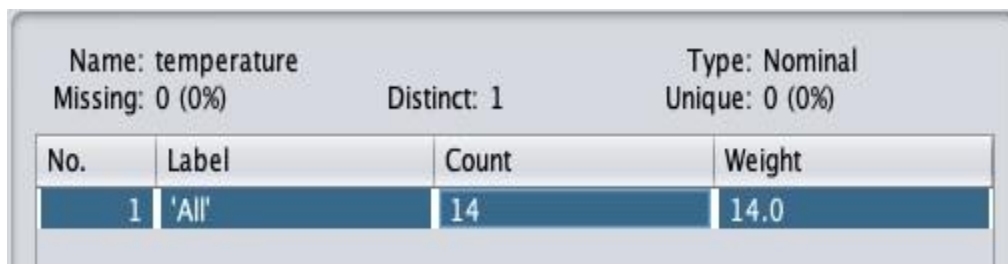
Some of the machine learning techniques such as association rule mining requires categorical data.

To illustrate the use of filters, we will use **weather.numeric.arff** database that contains two **numeric** attributes - **temperature** and **humidity**.

We will convert these to **nominal** by applying a filter on our raw data. Click on the **Choose** button in the **Filter** sub window and select the following filter:

weka→filters→supervised→attribute→Discretize

Click on the **Apply** button and examine the **temperature** and/or **humidity** attribute. You will notice that these have changed from numeric to nominal types.



The screenshot shows the 'Discretize' filter window in Weka. At the top, it displays 'Name: temperature', 'Missing: 0 (0%)', 'Distinct: 1', 'Type: Nominal', and 'Unique: 0 (0%)'. Below this is a table with four columns: 'No.', 'Label', 'Count', and 'Weight'.

No.	Label	Count	Weight
1	'All'	14	14.0

Supervised Filters

Definition: Supervised filters **USE the class label** to guide the transformation.

Key Idea: “Transform the data based on how it affects prediction of the target.”

Common Examples in Weka:

- Supervised → Attribute → AttributeSelection
- Supervised → Attribute → Discretize
- Supervised → Instance → Resample

Example:

Using the **Iris** dataset:

- Apply **AttributeSelection (InfoGain + Ranker)**
- Weka evaluates each feature based on how well it predicts the class (flower type)

Output:

- Features are ranked (e.g., petal length may be most important)

When to Use:

- Feature selection
- Improving classification accuracy
- Reducing dimensionality based on predictive power

Unsupervised Filters

Definition: Unsupervised filters **do NOT use the class label**.

They operate only on the input features (attributes), regardless of the output class.

Key Idea: “Transform the data without knowing the target.”

Common Examples in Weka:

- Unsupervised → Attribute → Normalize
- Unsupervised → Attribute → Standardize
- Unsupervised → Attribute → Remove
- Unsupervised → Attribute → Discretize
- Unsupervised → Attribute → ReplaceMissingValues

Example:

Using the **Iris** dataset:

- Apply **Normalize**
- Weka scales all numeric attributes to [0,1]

Important:

- It does **not care** whether the class is *setosa*, *versicolor*, or *virginica*

When to Use:

- Data cleaning
 - Scaling features
 - Removing attributes
 - Handling missing values
-